Taylor Polinomial

Section 9.7

3-3-26

A) Idea

e^x, $\sin x$, $\arctan x$, $\ln x$

write a 4th degree Taylor Polynomial of e^x created at 0.

$f(x) = e^x \rightarrow f(0) = 1$

$f'(x) = e^x \rightarrow f'(0) = 1$

$f''(x) = e^x \rightarrow f''(0) = 1$

$f'''(x) = e^x \rightarrow f'''(0) = 1$

$f''''(x) = e^x \rightarrow f''''(0) = 1$

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f^{n}(0)x^n}{n!}$$

$$P_4(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$$

Ex: Find the Taylor polynomial of degree n=4 for $f(x) = e^{3x}$ at 2

$f(x) = e^{3x} \longrightarrow f(2) = e^6$

$f'(x) = 3e^{3x} \longrightarrow f'(2) = 3e^6$

$f''(x) = 3^2e^{3x} \longrightarrow f''(2) = 3^2e^6$

$f'''(x) = 3^3e^{3x} \longrightarrow f'''(2) = 3^3e^6$

$f''''(x) = 3^4e^{3x} \longrightarrow f''''(2) = 3^4e^6$

$$P_4(x) = e^6 + 3e^6(x-2) + \frac{9e^6(x-2)^2}{2!} + \frac{27e^6(x-2)^3}{3!} + \frac{81e^6(x-2)^4}{4!}$$

Ex: Find the Taylor polynomial of degree n=4 for $f(x) = e^{3x}$ at 2.

Taylor Polynomial $c=0$

Maclaurin

Ex: Suppose that f is a function which has cont. deriv. and that

$f(2)=3, f'(2)=-4, f''(2)=7, f'''(2)=-5$

write the T. poly for degree 3 centered at 2.

$$P_3(x) = 3 - 4(x-2) + \frac{7(x-2)^2}{2!} - \frac{5(x-2)^3}{3!}$$

Find Maclaurin Polynomial of degree n=5 for $f(x) = \sin(x)$

$f(x) = \sin x \rightarrow f(0) = 0$

$f'(x) = \cos x \rightarrow f'(0) = 1$

$f''(x) = -\sin x \rightarrow f''(0) = 0$

$f'''(x) = -\cos x \rightarrow f'''(0) = -1$

$f''''(x) = \sin x \rightarrow f''''(0) = 0$

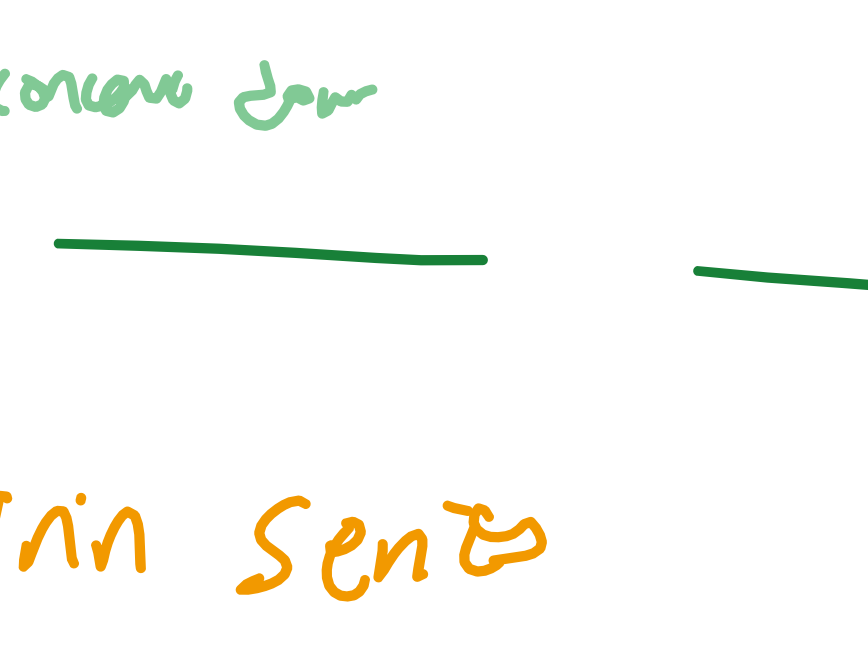
$f'''''(x) = \cos x \rightarrow f'''''(0) = 1$

$$P_n(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \ldots + \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

find $P_5(1.4)$

$$P_5(1.4) = 1.4 - \frac{(1.4)^3}{3!}$$

$$P_2(x) = a + bx + cx^2$$



$a = f(0) > 0$

$b = f'(0) > 0$ b/c f is increasing

$c = \frac{f''(0)}{2!} < 0$ b/c f is concave down

B) MacLaurin series

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \ldots + \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots + \frac{(-1)^n x^{2n}}{(2n)!}$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots + \frac{x^n}{(n!)}$$

C) Review

Ex: $\sin(x^2)$, $|error| < \frac{1}{1000}$, $\int_0^1 \sin(x)\,dx$

$$\sin x^2 = x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} + \frac{x^{14}}{7!} + \ldots + \frac{(-1)^n x^{4n+2}}{(2n+1)!}$$

$$\int_0^1 \sin x^2 = \frac{1}{3}x^3 - \frac{x^7}{7 \cdot 3!} + \frac{x^{11}}{11 \cdot 5!} - \frac{x^{15}}{15 \cdot 7!} + \ldots + \frac{(-1)^n x^{4n+3}}{(4n+3)(2n+1)!}$$

$$\approx \boxed{\frac{1}{3} - \frac{1}{7 \cdot 3!}}$$

alt

$\lim_{n\to\infty} a_n = 0$

decreasing

$|error| < |next\ term|$

$|next\ term| = \frac{1}{11 \cdot 5!}$

$$\lim_{x\to 0} \frac{\cancel{1} + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots \cancel{-1}}{2x} =$$

$$\lim_{x\to 0} \frac{1}{2} + \frac{x}{4} + \frac{x^2}{12} + \ldots = \boxed{\frac{1}{2}}$$

D) Geometric

$$S = \frac{a}{1-r} = a(r)^n$$

*a doesn't have to be constant

*If you want a certain x to be included in the IOC, you can shift the center.

$$f(x) = \frac{1}{1-2x} = 1 + 2x + 4x^2 + 8x^3 + \ldots (2x)^n + \ldots$$

$|r| = |2x| < 1$

$|x| < \frac{1}{2}$

$-\frac{1}{2} < x < \frac{1}{2}$

$$f(x) = \frac{1}{1-x^2} = 1 + x^2 + x^4 + x^6 \ldots + x^{2n}$$

$|x^2| < 1$

$|x| < 1$

$-1 < x < 1$

$$f(x) = \frac{12}{4+x} = \frac{3}{1-(-\frac{x}{4})} =$$

$$= 3 - \frac{3x}{4} + \frac{3x^2}{16} - \frac{3x^3}{64} + \ldots + 3\left(-\frac{x}{4}\right)^n$$

$\left|-\frac{x}{4}\right| < 1$ $|x| < 4$

$-4 < x < 4$

$f(x) = \frac{5}{4-x}$ @ $x=3$

$$= \frac{5}{4-(x-3)} = \frac{5}{1-(x-3)}$$

$$f(x) = 5 + 5(x-3) + 5(x-3)^2 + 5(x-3)^3 + \ldots + 5(x-3)^n$$

$|x-3| < 1$

$2 < x < 4$

$$f(x) = \frac{24}{3-(x-1)} = \frac{8}{1-\left(\frac{x-1}{3}\right)}$$

$$f(x) = 8 + \frac{8(x-1)}{3} + \frac{8(x-1)^2}{9} + \frac{8(x-1)^3}{27} + 8\left(\frac{x-1}{3}\right)^n$$

$\left|\frac{x-1}{3}\right| < 1$ $|x-1| < 3$

$\boxed{-2 < x < 4}$